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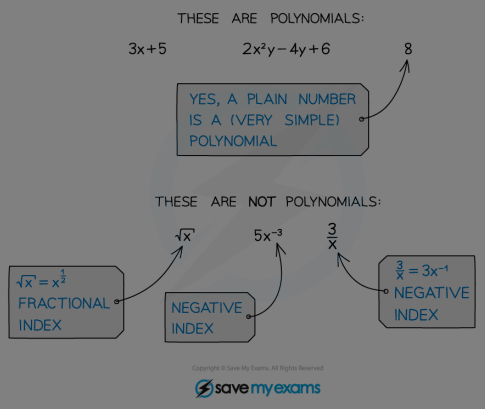
Factor theorem

What is the factor theorem?

- The factor theorem is a very useful result about polynomials
- A **polynomial** is an algebraic expression consisting of a finite number of terms, with non-negative integer indices only



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$$h(x) = 4x^3 + 8x^2 - 9x - 18$$
$$= (x+2)(4x^2 - 9)$$

BY (ii) THE FACTOR THEOREM TELLS US THAT h(−2) = 0 (TEST IT OUT!)

(x+2) = (x−(−2)) IS A FACTOR

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Examiner Tips and Tricks

- In an exam, the values of ***p*** you need to find that make **f(*p*) = 0** are going to be integers close to zero.
  - Try ***p*** = 1 and -1 first, then 2 and -2, then 3 and -3
  - It is very unlikely that you'll have to go beyond that.

Worked Example

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Expertise: Maths Content Creator

Roger's teaching experience stretches all the way back to 1992, and in that time he has taught students at all levels between Year 7 and university undergraduate. Having conducted and published postgraduate research into the mathematical theory behind quantum computing, he is more than confident in dealing with mathematics at any level the exam boards might throw at you.

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